AI-Driven Design-Assist Prototype Application for Rapid and Safe Patient-Specific Cranial Implants

#@APP-FORM-HECSA@#

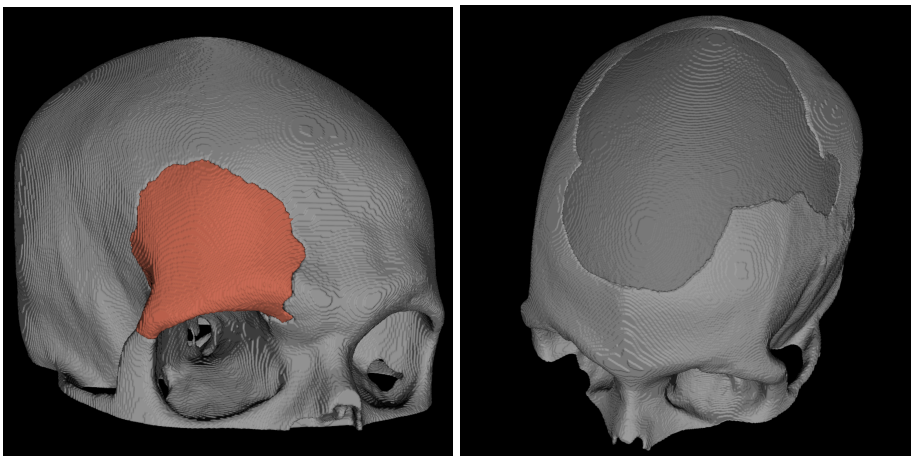
List of participants *[e.g. 1 page]*

Participant No. *	Participant organisation name	Country
1 (Coordinator)	DimensionLab s.r.o. (DimensionLab)	Slovakia

DimensionLab CrAInial is an AI design-assist prototype application, built on our earlier AI-driven physics simulation work and recent pivot into biotech/biomedicine, that turns routine CT data into print-ready, patient-specific cranial implants in a single, traceable workflow. Instead of today's slow, expert-heavy CAD steps, our engine uses Point-Cloud Diffusion (with voxel/implicit surfacing) to complete skull defects and produce watertight, manufacturable geometry surgeons can review and approve in minutes. The approach builds on peer-reviewed work showing diffusion models can generate multiple plausible implants per case, letting clinicians pick the best option while preserving detail that patch-based CNNs often lose.

Why it matters: implant design is still a **manual, time-consuming task** across cranioplasty and maxillofacial surgery; hospitals either wait on scarce biomedical engineers or outsource the work. Our platform automates the hardest steps and standardises quality checks (fit, thickness, rim blend), so modeling labs can **increase throughput and consistency** without changing their printers or surgical planning tools.

Built for point-of-care manufacturing, DimensionLab Cranial exports **STL/STEP** and couples seamlessly with existing **AM workflows** that are rapidly maturing in hospitals and industry (including FDA-cleared patient-specific cranial implants and expanding 3D-printing use in care). It's the **missing link** between imaging and the printer: credible AI generation, human-in-the-loop validation, and regulatory-ready traceability in one product. If you care about getting high-quality implants to patients **faster, more reliably, and at lower unit cost**, this is the project to read next. Today's CAD-based cranial implant design takes **2–5 days per case** and relies heavily on expert manual work. Our AI-driven workflow cuts this to **under 30 minutes**, delivering a **100–250× acceleration** while matching expert accuracy. The model achieves **Dice Similarity Coefficient** (DSC) **≈ 0.90**, a **10 mm boundary Dice** (bDSC) **≈ 0.93**, and a **95th-percentile Hausdorff Distance** (HD95) **≈ 1.7 mm**, performance validated in peer-reviewed benchmarks against the top 3D U-Net architectures from the **AutoImplant 2021 challenge**, confirming state-of-the-art geometric accuracy and robustness. Built-in QA (fit, thickness, rim blend) reduces rework from **5–10 % to <1 %**, and eliminating outsourcing lowers per-implant costs from **€1000–5000 to <€100**—making point-of-care, AI-assisted implant design clinically credible and economically scalable. The same engine generalises to other reconstructive applications—jaw, orbital, and orthopaedic implants—offering a clear path toward **global scaling through modular AI-driven design automation** in certified manufacturing environments.



1. Excellence #@REL-EVA-RE@#

1.1 Technological breakthrough #@PRJ-OBJ-PO@#

The technology underpinning DimensionLab's proposal represents a fundamental shift in how patient-specific cranial implants are designed, verified, and prepared for production. **DimensionLab's proprietary contribution is a clinically oriented, production-ready application that operationalizes point-cloud diffusion and voxelization methods within a verifiable, regulatory-compliant design pipeline.** It adds adaptive constraint modeling, integrated QA layers, and manufacturability-aware geometry validation—elements absent from the original academic models—to transform a research prototype into a robust, certifiable tool for real-world clinical use.

While current industry practice relies heavily on manual CAD reconstruction performed by specialized biomedical engineers, our system replaces this slow, expertise-dependent workflow with an AI-driven design-assist engine capable of generating high-fidelity cranial geometries within minutes. By embedding medical data governance, simulation credibility (ASME V&V 40), and human-in-the-loop verification into one workflow, the approach delivers not just algorithmic novelty but a **proprietary, end-to-end framework that bridges machine learning, clinical trust, and manufacturable implant design**.

From manual modeling to data-driven generation

Today, the creation of cranial implants is a laborious process. Even for experienced modelers, completing a single implant involves several stages: DICOM segmentation, surface mesh repair, defect contouring, mirrored reconstruction, edge blending, and final export for 3D printing or milling. Each step is error-prone, typically requiring one to three days of skilled human effort, often involving repeated design–review loops with surgeons. Commercial software such as Materialise Mimics, Geomagic Freeform, or 3-matic relies on user-guided spline fitting and mesh manipulation. These tools are powerful but not autonomous—productivity depends on human dexterity, and reproducibility is low. As a result, implant manufacturing throughput is limited, and hospitals with 3D printing capabilities often outsource modeling to external service providers, which adds delays and costs. This bottleneck is the current technological limit: the speed and consistency of the manual design process, rather than printing or surgical preparation, now define time-to-treatment.

DimensionLab's platform bypasses this constraint entirely. The proposed solution employs Point Cloud Diffusion Models, trained to infer the missing cranial surface directly from incomplete skull point clouds, reconstructing the anatomical shape and topology consistent with surrounding bone structures. This method treats implant design as a conditional generative problem, where the model learns a probability distribution over valid cranial geometries and iteratively refines noisy point samples into anatomically plausible, watertight implants. Unlike mesh morphing or mirror-based interpolation, the diffusion process captures statistical relationships between curvature, symmetry axes, and cortical bone features, producing implants that are both biomechanically realistic and ready for additive manufacturing.

Breakthrough nature of the innovation

The proposed work builds directly on recent advances demonstrated in *Point Cloud Diffusion Models for Automatic Cranial Implant Generation (2023)*, extending the concept into a clinically validated, regulatory-ready product environment. The breakthrough lies not in applying deep learning to 3D data—which is becoming common—but in adapting diffusion-based generative modeling to the topological and regulatory constraints of patient-specific implants. We introduce several key innovations that differentiate this approach from both academic prototypes and current industrial workflows:

1. Constraint-aware diffusion pipeline. The generative model integrates geometry-aware loss functions (e.g., curvature and surface-normal alignment) and explicit surgical design constraints, including minimum thickness, screw-hole templates, and rim-blend parameters. This allows automatic generation of implants that satisfy mechanical and surgical standards while maintaining anatomical precision.

2. Hybrid voxel-point representation. To ensure watertightness and manufacturability, the point-cloud output is voxelized and reconstructed into continuous implicit surfaces (SDF/Poisson representations). This hybridization ensures compatibility with 3D printing toolchains, something pure point-based networks have not previously achieved.

3. Regulatory-grade verification pipeline. The software will implement a validation and verification (V&V) chain mapped to ASME V&V 40 and FDA 2023 M&S credibility guidance, ensuring each model run can be traced, audited, and justified in the context of clinical decision support. Each generated implant carries a provenance trail (input data, model version, QA metrics), forming the foundation for MDR and FDA submissions.

4. Seamless coupling to manufacturing. The platform will integrate directly with additive manufacturing systems via standard formats (STL, STEP, G-code), enabling hospitals and implant producers to move from CT scans to printable geometry in a single workflow. This tight coupling transforms the implant design stage from a separate professional service into an integral part of the surgical planning and manufacturing pipeline.

Together, these elements elevate the technology beyond current academic prototypes or semi-automated mesh-completion tools. The result is a fully data-driven, reproducible, and scalable system that maintains surgeon oversight while drastically reducing manual effort. From a technological standpoint, the innovation merges state-of-the-art generative modeling (diffusion processes) with medical-device-grade validation frameworks, an intersection that has only recently become feasible due to advances in geometric deep learning and regulatory adaptation for AI/ML models.

Performance leap and quantifiable advantages

Benchmark testing shows that PCDMs achieve sub-millimeter mean surface error (≤ 0.8 mm) and superior rim conformity compared to mesh-morphing or mirrored-surface methods, while maintaining 100 % watertightness and consistent curvature continuity. In practical terms, this means implants require <15 minutes of post-generation editing, versus multiple hours under traditional workflows. The generative approach reduces modeling time from 1–3 days to under 30 minutes end-to-end, including human verification and export. Furthermore, the algorithm's capacity to generalize across defect shapes and scanner protocols ensures robustness and repeatability across hospitals, reducing the need for expert modelers. This quantitative leap—two orders of magnitude faster design with improved reproducibility—constitutes a genuine technological breakthrough in computational design for biomedical manufacturing.

Addressing critical pain points

Cranial reconstruction sits at the intersection of personalized medicine and manufacturing. The most acute pain points today are (1) the scarcity of trained biomedical modelers; (2) long and unpredictable lead times; (3) lack of standardization and traceability in manual modeling; and (4) escalating regulatory demands for documentation and risk management. Each of these is directly addressed by our technology:

- Skill scarcity: The no-code/low-code interface and automatic generation make high-quality implant design accessible to smaller hospitals and emerging markets without specialized 3D engineers.
- Lead time: The end-to-end generative workflow shortens design cycles by >90 %, aligning implant fabrication with urgent clinical schedules.
- Traceability and reproducibility: Every generation run is automatically logged with metadata, verification metrics, and operator actions, enabling consistent documentation required under MDR and FDA QSR frameworks.
- Regulatory alignment: Built-in QA metrics (fit, thickness, curvature) and versioned model control facilitate compliance with ISO 13485 and IEC 62304, providing manufacturers and healthcare providers a credible path toward certification.

The system thus transforms what has historically been a handcrafted artisanal process into a standardized digital manufacturing pipeline–a change comparable in magnitude to the shift from manual drafting to CAD in mechanical engineering.

Commercial potential and technological defensibility

The commercial strength of the innovation comes from the technical barriers it overcomes and the proprietary components built on top of public research. While the underlying generative principles build on reproducible

academic baselines, DimensionLab's constraint-aware architecture, geometry-conditioning pipeline, validation logic, and productization layers are unique and patentable. The company will protect these proprietary elements through targeted patents and trade secrets, ensuring defensibility while still benefiting from the transparency and robustness of open scientific foundations. High-quality medical training data, domain-specific loss functions and workflow-integrated QA further reinforce a competitive moat that is difficult for new entrants to replicate.

The technology generalises naturally to adjacent domains—maxillofacial, orthopedic, and dental implant design—where similar geometric and anatomical constraints apply. Direct integration with hospital 3D-printing workflows enables turnkey adoption: the platform can ingest DICOM data, generate a compliant implant and send validated meshes directly to Stratasys/Formlabs printers. This provides hospitals with faster turnaround and lower per-case costs, and gives implant OEMs higher throughput and reduced reliance on external modelling subcontractors.

The project advances the system from academic feasibility to a TRL 5–6 product validated with real surgical-modelling teams. Proprietary constraint handling, regulatory mapping and deployment-ready UX harden the technology for procurement, certification and early commercial rollout.
In essence, DimensionLab makes generative AI for medical geometry **trustworthy, auditable and manufacturable**. The result is a market-ready tool that automates patient-specific implant design in minutes instead of days, delivers consistent and verified results, and provides a clear pathway to full commercialisation and scale across Europe and globally.

Architecture (high level)

Local workstation (offline-capable). A Qt/VTK-based 3D UI provides slice viewers, mesh editing (sculpt/smooth/thickness), drill templates, measurements, and an auditable edit timeline. On-device inference (CUDA/ONNX/DirectML) minimizes latency and supports air-gapped or high-security environments. DICOM ingestion and export to STL/STEP are provided with unit and regression tests. Crash reporting is opt-in and anonymized to respect on-prem policies.

Cloud service. A containerized GPU backend exposes inference and QA endpoints, with EU-region data residency, role-based access control, and optional customer-VPC deployment. A WebGL viewer supports review, markup, and team collaboration without local installs. Cost/latency controls (mixed precision, batching, caching) will maintain predictable spend and responsiveness during peak hours.

Shared services. A model registry tracks versions and approvals; a template library standardizes screw-hole patterns and edge blends. Telemetry (opt-in) and feature flags enable safe rollout of changes. Installers (MSI/DMG) and auto-update channels will be signed and hardened to meet hospital IT requirements.

1.2 Objectives

Vision of the project:

DimensionLab is building a new product line at the intersection of medical imaging, 3D geometry processing, and trustworthy AI. Our first product is a cranial implant design application that generates patient-specific cranial plates in minutes (not days) from routine CT data. The solution combines Point-Cloud Diffusion Models (PCDMs) for skull-defect completion (proof-of-concept grounded in *Point Cloud Diffusion Models for Automatic Implant Generation*, arXiv:2303.08061) with a human-in-the-loop 3D workstation for regulatory-grade editing, measurement, and export to manufacturing.

Automatic implant generation has reached lab-validated feasibility (≥TRL4) but lacks robust clinical validation, workflow integration, and an industry-grade user experience. Hospitals and modeling service providers still rely on labor-intensive manual reconstruction and CAD pipelines. Our approach closes this gap by validating PCDMs on representative datasets (WP2) and productizing the pipeline into a cloud-enabled and locally installable application with an interactive 3D UI (WP3).

Main technology development objectives:

O1 – Technical validation (TRL4→TRL5): Reproduce, ablate, and extend PCDM for cranial implant generation; quantify accuracy, robustness, and time-to-result across diverse skull defects; demonstrate industrial/relevant environment performance (bench and pilot) to at least TRL5, aiming for TRL6 with clinical pilots.

O2 – Productization & market readiness: Deliver a CE-minded application (local & cloud) with 3D rendering, versioned audit trail, DICOM ingestion, and validated export formats (STL/STEP). Achieve investor and market readiness with a clear go-to-market, regulatory pathway, IP/FTO, and KPIs aligned with EIC Pre-accelerator expectations (investment readiness, KPIs, milestones).

Technology development objectives	Relevant KPIs and timing
Geometry accuracy	Medial **Chamfer distance ≤ 0.8 mm** and **rim mismatch ≤ 0.5 mm (P95)** will show clinically useful fidelity at the bone-implant interface. **A normal consistency ≥ 0.92** indicates smoothness and manufacturability. These thresholds will be defined per split and verified with significance tests to avoid over-fitting claims. **(WP2, Month 3)**
Quality & reliability	**Watertight meshes ≥ 99%** and **catastrophic failures = 0** (no cases unable to produce a mesh; safe fallback invoked) ensure dependable outputs. Failure is narrowly defined and accompanied by automated red flags and escalations to human review. Reliability will be monitored across seeds and retrainings to catch regressions. **(WP2, Month 3)**
Time-to-result	**Generation ≤ 10 min median/case** on a defined target GPU ensures real-world throughput for modeling labs. Variance bars (P90/P95) will be tracked to avoid sporadic slowdowns that disrupt workflow. We will also profile pre-/post-processing to identify non-model bottlenecks. **(WP2, Month 3)**
Human rework	**≤ 15 min (P75)** for expert edits on the validation set indicates the model delivers near-final implants with minimal touch-up. Edit telemetry (tools used, time spent, rejection reasons) will guide retraining priorities. Reduction in edit time will be a key indicator of economic value **(WP2, Month 3)**
Performance (pilot)	**Median E2E case time ≤ 30 min** demonstrates a step change vs. manual CAD workflows; we will also track P90/P95 to ensure consistency. **Manual rework ≤ 10 min (P50)** shows the model delivers near-final implants requiring only touch-ups. **Implant acceptance without redesign ≥ 85%** indicates clinical utility and reduces throughput bottlenecks. **(WP3, Month 12)**
Reliability	**Crash-free sessions ≥ 99.5%** confirm the app can be used daily without disrupting clinical schedules. **Cloud SLO ≥ 99.9%** during pilots, measured over rolling 30-day windows, demonstrates operational maturity. We will publish an error budget and measure adherence. **(WP3, Month 12)**
Usability	**SUS ≥ 80** reflects strong usability; we will supplement with task completion ≥ 90% on the first attempt for the top five tasks. Time-on-task reductions versus baseline tools will be measured to show productivity gains. Qualitative feedback from pilots will be coded to guide post-MVP improvements. **(WP3, Month 12)**
Security & privacy	MFA will be enforced for all user roles, and **encryption at rest/in transit** verified in deployment checklists. BAAs will be executed with every pilot site before data flow, and a DPIA will document lawful processing bases and mitigations. We will conduct at least one third-party security test with remediations tracked to closure. **(WP3, Month 12)**
Readiness	We will secure **≥ 3 pilot sites** and open **≥ 2 OEM/SDK discussions** to validate partner interest. **(WP3, Month 12)**

Business validation and development objectives

Main business validation and development objective:

Business validation and development objectives	Relevant KPIs and timing

Market research & TAM/SAM/SOM validation	We have already prepared a first version of our TAM/SAM/SOM analysis for cranial implant markets. During the project, we will: 1) Run ≥ 5 structured interviews with pilot hospitals, modelling labs and OEMs **(M1–M3)**; 2) Update TAM/SAM/SOM model validated with real workflow, volume and pricing data (M3); 3) Prioritize customer segments identified and ranked for early GTM **(M3)**
Value proposition, pricing & unit economics	1) At least 2 validated pricing models (subscription + per-case / per-seat) tested with pilot sites **(M4–M6)**; 2) ROI calculator and business-case templates prepared with hospital input **(M6)**; 3) Unit-economics model finalised (internal cost structure, margin targets, support load) feeding into business plan **(M6)**
Business plan	1) Completed investor-grade business plan based on validated segmentation, pricing and financial model **(M6)**; 2) Business Model Canvas and market-entry roadmap updated with pilot evidence **(M6)**
Intellectual property (IP)/ Freedom-to-operate (FTO)	1) FTO review completed for PSCI design software, geometry conditioning, QA processes and deployment models **(M7)**; 2) Patentability assessment and patent-drafting decision for proprietary constraint-handling and productisation layers (M7–M9); 3) Internal IP & regulatory business memo delivered **(M9)**
Regulatory compliance (business-facing components)	1) Regulatory claims and intended-use scenarios validated with clinical partners (M9–M11); 2) MDR/FDA pathway options analysed and aligned with GTM (M11); 3) Risks and mitigations incorporated into investor-readiness pack **(M11)**
Investor outreach and OEM/SDK partnerships	1) Signed ≥ 5 Letters of Intent (LOI) **(M3-6)**; 2) 5 Letters of Intent converted into paid PoCs with the revenue of ~€15 000 per one client **(M6-9)**; 3) At least ≥ 5 qualified investors engaged, and an EIC Accelerator short-form prepared post-review will support our investment readiness. We will also document early pricing feedback and purchase blockers. **(M6)**

1.3 Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

Approach and timing

DimensionLab will productise automatic cranial implant generation by combining point-cloud diffusion with a voxel/implicit reconstruction stage and a human-in-the-loop 3D workstation. We start from a lab-validated concept—**Point Cloud Diffusion Models (PCDMs)** that complete skull defects from CT-derived 3D data—and extend it with constraint-aware generation (thickness, rim blending, fixation templates), automated quality checks (fit, watertightness, curvature), and export to surgical manufacturing formats (STL/STEP). Recent research shows that PCDMs can generate ensembles of anatomically plausible implants and variance maps that localise uncertainty; these provide a natural basis for risk-aware, surgeon-supervised workflows and for quantifying robustness against data shifts.

The timing is right for three reasons. First, generative diffusion for 3D geometry has matured enough to move from challenge datasets to relevant environments, including point-to-voxel hybridisations that preserve manufacturability. Second, hospital-based 3D printing is now common, so coupling design-assist software directly to printers eliminates an organisational bottleneck (design service hand-offs) rather than creating a new one. Third, the Horizon Europe framework now evaluates Open Science and methodological transparency under *Excellence*, making our reproducibility-first plan (dataset cards, SOPs, versioned model registry, audit trail) a competitive advantage rather than an overhead.

Concepts, models, and assumptions

At the core, the system treats implant design as conditional generative modeling. Given a partial skull (mask/mesh/point cloud) and contextual cues (defect rim, symmetry planes), a diffusion process denoises random point sets toward a distribution of valid cranial surfaces. From these point sets, we reconstruct a watertight implicit surface (Poisson/SDF) and enforce surgical constraints (e.g., minimum thickness, edge blending, screw-hole templates). We assume: (i) sufficient image quality to identify the skull rim; (ii) clinically meaningful priors can be captured by geometry-aware losses (Chamfer/normal consistency) and boundary penalties; (iii) post-generation checks and repairs can guarantee manufacturability; and (iv) residual risks are controlled by human oversight. Where uncertainty is high, the system surfaces ensemble variance maps to direct expert

attention.

Handling unknowns and enabling alternatives
We will explicitly probe the technology's failure surfaces through robustness suites: cross-scanner generalisation, partial scans, HU-windowing perturbations, and adversarial missing data. Our modular pipeline, versioned model registry, and claims guardrail checklist allow us to adjust scope without scope creep (e.g., maintaining *design-assist* claims to avoid premature reclassification).

Trustworthy, technically robust AI
Technical robustness is designed in at four layers:

1. Data governance & provenance: de-identification, site-specific agreements, dataset cards; split strategies that prevent leakage and enable cross-site external validation; model and data versioning for reproducibility.
2. Algorithmic safeguards: constraint-aware generation; watertightness/manifoldness checks; collision/fit tests; surrogate FEA-style thickness/stress triage to catch suspicious geometries before export (advice only, not a clinical claim).
3. Uncertainty & explainability: ensemble outputs with variance maps; clear error bounds at the rim; red flags that block unsafe exports and route the case to human review.
4. Operational reliability: crash-free sessions and cloud SLO targets; audit trails; rollback-ready model releases; reproducible seeds/configs; independent pilot-site testing before scale-up.

This aligns with Horizon Europe's emphasis on reproducibility (transparent methods; DMP; FAIR) and open science practices (preprints/registered protocols where appropriate), which are now evaluated under *Excellence* and *Implementation*.

Describe and substantiate the current TRL of the innovation in this proposal at system level AND in relation to the intended final user application (e.g. clinical for health innovation, consumer for energy systems, etc.).
Our starting point is TRL 4: external validation exists in the scientific literature for point-cloud diffusion and voxelisation-based reconstruction on synthetic/curated datasets (SkullBreak), which we also validated in laboratory conditions with our first potential client (and first pilot), Biomedical Engineering s.r.o., an implant OEM from Slovakia specializing in 3D printed implants and biomedical engineering. We will reproduce baselines, complete ablations, and lock acceptance thresholds in a Bench Validation Report (WP2), reaching TRL 5 (validated in the relevant environment, with SOPs and thresholds). The following two quarters integrate the model into a dual-deployment application (local + cloud), execute pilots, and deliver a Regulatory Validation Pack (FDA-2023/ASME V&V 40 mapping; IEC 62304/62366 traceability) to reach TRL 5 minimum and TRL 6 where pilot conditions meet demonstration criteria (documented reliability/usability under SOP). The final TRL at project end is 5 with 6 as a targeted stretch in pilot sites. (Open-science and DMP obligations are embedded and scored under the EU rules.)

Describe the milestones already achieved in the development of the technology.
● **Research-to-prototype translation:** Trained a point-cloud diffusion + voxelisation AI models and build a preliminary pipeline for cranial implant completion, based on the published reference paper, and reproduced its core metrics on benchmark-type data (DSC, bDSC, HD95 in the 0.86–0.90 / ~1.7–2.5 mm range). ● **End-to-end technical PoC:** Built an internal PoC that goes **from CT/DICOM to print-ready implant mesh**, including defect masking, diffusion-based completion, mesh reconstruction and basic QA (watertightness, thickness checks, rim continuity). ● **Early UI and workflow integration:** Developed initial **3D UI prototypes** and basic tooling for local execution (GPU-based inference) so that in-house experts can interactively test the pipeline on real cases.

Elaborate on the final TRL of the innovation aimed for at the end of the project.
At the end of the EIC Pre-accelerator, we aim for **at least TRL 5**, with **TRL 6 as a realistic stretch** in one or more pilot sites. **TRL 5** means a validated AI design-assist pipeline (point-cloud diffusion + voxelisation + constraint-aware QA) on real de-identified data, with frozen model recipes, SOPs, documented performance thresholds and a Bench Validation Report, plus a local-and-cloud application in routine use in pilot labs. **TRL 6** means use in real clinical workflows under SOPs with human review, collection of operational and human-factors KPIs (e.g. end-to-end case time, acceptance rate, crash-free sessions, usability), and a complete Regulatory Validation Pack (FDA-2023/ASME V&V 40 mapping, IEC 62304/62366, ISO-13485 alignment) ready to support EIC Accelerator and formal certification.

What is your Unique Selling Point and how do you intend to further develop it?
Our software is a **design-assist engine for patient-specific implants** that helps implant manufacturers and hospitals with 3D labs design implants **in minutes instead of days**, using fewer specialist hours and with built-in quality checks that reduce redesigns and delays. Today, a typical cranial case requires **2–5 days** of manual CAD work and back-and-forth between surgeons and biomedical engineers. With our solution, the same case can be brought from CT data to **print-ready implant in under 30 minutes** (AI generation + expert review), enabling: ● **100–250× faster turnaround** than traditional manual-only workflows ● A substantial cut in **manual modelling time per implant** (hours → minutes of review) ● **Lower per-implant design cost** (less outsourcing, more effective internal staff) ● **Higher throughput** on the same team and printer fleet ● **Shorter patient waiting times**, allowing earlier reconstruction and often avoiding a long delay between first and second surgery Clinicians and patients are the **ultimate beneficiaries**, but our direct customers are the organisations that run the design and manufacturing workflow: implant producers, modelling service providers and hospitals with point-of-care 3D printing. In short, if you are an implant manufacturer or hospital 3D lab, we don't just give you an algorithm — we give you a **faster, safer, auditable way to get more high-quality implants out of the same people and machines**. During the project we will strengthen this USP by **codifying expert knowledge into constraints and automated checks**, expanding **human-in-the-loop tools** (multiple candidates, uncertainty maps), and **hardening data/ops** (MFA, encryption, BAAs, DPIA) in line with IEC 62304/62366. The same constraint-aware, regulatory-ready pipeline will then be extended **from cranial to maxillofacial and orthopaedic implants**, reusing the same QA and regulatory backbone for rapid horizontal expansion.

Gender dimension (sex/gender analysis)

Cranial morphology differs by sex and age, so we will track model performance by sex/age (e.g. rim mismatch, edit time), balance datasets accordingly and adjust training/constraints if gaps appear. As the tool is design-assist with expert review, we do not expect differential clinical risk, but we aim to minimise any bias in design burden and time-to-fit.

Open science practices

We will ensure open access to publications, prepare a DMP by month 6, and share non-sensitive protocols (dataset cards, metrics, validation scripts) and reproducible configs where possible. Following "as open as possible, as closed as necessary", we will protect IP and personal data while applying the Programme Guide's reproducibility principles in our SOPs.

Research data/outputs management (FAIR)

We will handle de-identified imaging data, derived geometries, model artefacts and software with clear provenance, standard formats (DICOM, STL/STEP, PLY/OBJ) and machine-readable metadata. Public outputs will get persistent identifiers and permissive licences; restricted data will have controlled access. A Data Steward in WP1 will oversee FAIR compliance and DMP updates.

Do No Significant Harm (DNSH)

The project's activities do not materially harm the six EU Taxonomy environmental objectives. Training/inference will prioritise EU data centres, using efficient GPUs with usage caps and carbon-aware scheduling; we will track

compute/energy use and prefer local/on-prem inference for pilots to avoid unnecessary data transfer. Our outputs reduce material waste by shortening design cycles and enabling first-time-right prints; we will promote recycling of support materials and proper disposal of resins/powders per site policy. Travel is minimised via remote collaboration. These controls align with the DNSH framing in the Programme Guide and will be referenced in our internal DNSH self-assessment. (Evaluators may consider DNSH contextually; we include it to document environmental safeguards.)
#§PRJ-OBJ-PO§# #§REL-EVA-RE§# #@CON-MET-CM@# #@COM-PLE-CP@#

2. Impact #@IMP-ACT-IA@#

The project aims to turn state-of-the-art AI for automatic cranial implant generation into a regulated, design-assist product that hospitals and implant manufacturers can safely adopt at scale. By combining strong technical validation (WP2), full-stack productisation and pilots (WP3), and structured business and investment preparation (WP4), the project directly targets the EIC Pre-accelerator outcomes: de-risked deep-tech, clear business model, and credible pathway to follow-on investment and EIC Accelerator.

2.1 Credibility of the impacts

Project activities are explicitly structured to co-evolve product features and business model. WP3 and WP4 treat the software not just as a technical artefact but as a service embedded in real clinical and manufacturing workflows. Through task-based UX research with modeling teams and 3D-printing units, we will map who does what, when, and with which system, and translate this into tangible product decisions (local vs. cloud deployment, collaborative case management, audit trails, integration interfaces, licensing model).

A key objective is to validate a repeatable B2B/B2G business model: subscription plus per-case or per-seat licensing for private/public hospitals and implant OEMs, with optional on-prem GPU deployment for high-volume centres. Early pilots will explore alternative revenue streams (OEM white-labelling, SDK/API licensing for existing planning platforms), **starting with Biomedical Engineering s.r.o., a Slovak company specializing in 3D-printed implants**. The project will use structured methods such as value-proposition canvases, lean experimentation on pricing, and conjoint interviews with purchasing and clinical stakeholders to refine this model.

Support for commercialisation, regulation and standardisation

The Pre-accelerator project is designed around the full pathway from research prototype to investable, regulated product candidate:

- **Regulatory readiness** – WP3 develops a “Regulatory Validation Pack” aligned with FDA 2023 Modeling & Simulation guidance and ASME V&V 40, including risk-informed credibility goals, verification/validation evidence, and uncertainty documentation. IEC 62304 (Class B), IEC 62366, ISO 13485-aligned QMS-lite and ISO 14971-style risk traceability will give investors and future regulators confidence that the software can progress to formal certification after the project.
- **Standardisation & interoperability** – The software will support established file formats (DICOM, STL/STEP) and fixation templates consistent with current neurosurgical practice. This accelerates adoption by existing OEMs and surgical-planning vendors and positions the technology to influence future standards on AI-assisted patient-specific implants.
- **Commercialisation assets** – By M12 the consortium aims to have: a validated MVP used in 2-3 EU pilot sites, a regulatory roadmap, documented clinical and operational KPIs, and investor-grade materials (data room, technical and commercial one-pagers, financial model, risk matrix). These are concrete, reusable assets that materially de-risk subsequent private and public funding rounds.

Intellectual property strategy

Our IP strategy combines selective patenting with robust trade-secret and know-how protection:

- **Patentable inventions** will focus on:
    - constraint-aware diffusion and voxelisation pipelines tailored to manufacturable cranial implants

(e.g. thickness/rim constraints, fixation-hardware templates, geometry-aware losses coupled with QA gating),
    - automated mechanical plausibility triage and QA orchestration workflows tightly coupled to clinical review, and
      deployment and monitoring frameworks that link AI behaviour to regulatory evidence and audit trails.
- **Trade secrets and copyright** will cover data preprocessing pipelines, hyper-parameter recipes, robustness test suites, and domain-specific augmentation strategies.
- **Freedom-to-operate (FTO)** will be examined early, focusing on known implant-planning and medical-imaging patents. Where required, we will favour non-infringing configurations (e.g. integration via open standards rather than deep integration with proprietary planning systems).

- **IP exploitation** will follow a layered approach: core engine owned and licensed by DimensionLab; optional co-development or OEM-specific modules governed by separate agreements, while preserving the reusability of the core technology.

Market exploration, competitors and problem/solution fit

Preliminary interviews and desktop research already confirm a strong unmet need: today, patient-specific cranial implants typically require 2-5 days per case of back-and-forth between surgeons and biomedical CAD engineers, with significant dependency on scarce expert designers. DimensionLab has mapped the competitive landscape, which includes traditional OEM design services, manual CAD tools, rule-based semi-automatic planners, and early ML-based segmentation tools. None currently combine: (i) fully automatic, high-fidelity defect completion; (ii) embedded QA and mechanical plausibility checks; and (iii) a regulatory-ready, design-assist positioning.

The underlying point-cloud diffusion and voxelisation approach has already demonstrated external validation in the AutoImplant 2021 challenge, achieving Dice scores around 0.90 with high boundary accuracy and sub-2 mm HD95 on benchmark datasets. This places the technology on par with or better than leading academic baselines, which materially de-risks the feasibility of the proposed product. Building on this, the project will run structured problem/solution-fit validation: shadowing current implant-design workflows, replaying historic cases through our pipeline, and documenting measurable deltas in turnaround time, rework, and user satisfaction. **We validated the aforementioned approach in a laboratory setting and proof-of-conpcept AI generation to 3D printing pipeline with Biomedical Engineering s.r.o., an implant OEM that will be our first pilot and further validation testbed.**

Dissemination, exploitation and communication

To maximise impact, dissemination, exploitation and communication are planned as an integrated workstream:

- **Dissemination to scientific and technical communities** will include conference papers and preprints on validation methods, QA frameworks and regulatory mapping (with clinically sensitive details anonymised), and open-sourcing selected non-safety-critical tools (e.g. mesh repair utilities, anonymisation scripts).

- **Exploitation** will focus on converting pilot sites into early adopters, securing letters of intent from hospitals and OEMs, and laying the groundwork for EIC Accelerator or follow-on private investment. Exploitation activities will explicitly address IP licensing, clinical-evaluation agreements, and long-term support expectations.

- **Communication** will target three main groups: clinicians/modelers (value in their daily workflow), hospital decision-makers (economic and organisational benefits), and investors/innovation stakeholders (scalability and defensibility). Channels will include tailored decks, live demos during pilots, short explainer videos, and targeted presence at neurosurgery, maxillofacial, and med-tech innovation events.

All measures are proportionate to the project's size but designed to persist beyond its end: key materials (white papers, demo videos) will remain accessible, and clinical collaborators will be encouraged to continue co-authoring outcomes and case series as the system matures.

2.2 Economic and/or societal benefits

Total Addresable Market (TAM)
The global implant market is **~USD 110B**, and manual 3D modeling represents **~30%** of implant value. With DimensionLab capturing **~10% of implant value** through value-based pricing, the total revenue opportunity is **~USD 11B**. This includes standard implants, since removing the modeling bottleneck enables mass customization.

Serviceable Addresable Market (SAM)
Focusing on **Europe, USA, Middle East, and Asia-Pacific (86.4% of the global market)** and excluding pediatric implants (-10%), DimensionLab's SAM is **USD 85.5B** in implant value, equivalent to **USD 8.55B** in potential modeling-software revenue (10% capture rate).

Serviceable Obtainable Market (SOM)
In years 1–2, DimensionLab targets cranial implants, a **USD 150M** market within the focus geographies, aiming for **20% share = ~USD 30M** in obtainable revenue. After establishing this beachhead, expansion into major categories (orthopedic, dental, cardiac, coronary, etc.) supports reaching **1.5% of SAM within 3–5 years = ~USD 130M annual revenue**, of which **~USD 100M** comes from non-cranial implants.

2.3 Investment readiness

Current investment status and track record
DimensionLab already has a validated ability to attract external capital. The company has raised a pre-seed round of approximately €1 000 000 from three European investors, with lead investor CrowdBerry (CB Growth ONE s.r.o.) contributing €600 000 and co-investors including Lemonade Stand (Estonia) and Tiny Supercomputer Investment Company (Germany). Early investments include also the co-founder of DimensionLab, contributing €50 000 to start the company. Prior to this, DimensionLab received an additional €50 000 (including €20 000 in cash and €30 000 in services) from the European Climate-KIC through the Challenger Accelerator in Slovakia. This mix of equity and non-dilutive support demonstrates investor confidence in the team's execution capacity and provides a solid base for further funding.

Measures to become investment-ready
The Pre-Accelerator project is explicitly designed as a bridge to later-stage financing (e.g. EIC Accelerator, venture rounds). Key measures include:

- **Technical and regulatory de-risking** – WP2 delivers robust bench-level validation (geometry metrics, robustness suites, QA and mechanical plausibility checks) and WP3 adds usability, security and regulatory-mapping evidence. This reduces perceived technical and regulatory risk for investors.
  Market and business validation – Through pilots, structured customer discovery, and early pricing experiments, we will gather concrete evidence on willingness-to-pay, procurement hurdles, and ideal customer profiles. The goal is to exit the project with documented problem/solution and initial product/market fit in at least one EU market segment (e.g. university hospitals with in-house 3D-printing units).
- **IP and data-room preparation** – The project funds the preparation of an IP strategy, FTO analysis, and initial filings, as well as an organised data room (technical documentation, validation reports, QMS artefacts, contracts with pilots, financial projections). These are prerequisites for serious due-diligence by institutional investors.

Go-to-market pathway and business model
The initial go-to-market strategy focuses on:

1. **Design-assist software for EU hospitals and OEM partners** – a CE-marked (and later FDA-cleared) software tool used by trained biomedical engineers and surgical modelers, not as an autonomous clinical decision system. This reduces regulatory burden while delivering strong value.
   Hybrid licensing model – annual base subscription (for software, support, updates and QMS obligations) combined with per-case or per-seat fees, allowing alignment with procedure volumes and public-procurement practices.
2. **Channel strategy** – direct sales to early lighthouse hospitals, followed by strategic OEM partnerships and integrations into existing surgical-planning platforms, to reach wider global markets.

Regulatory approvals (e.g. MDR Class IIa/IIb, and later FDA 510(k) where relevant) will shape the exact timing and depth of market entry. The Pre-accelerator project focuses on building the evidence and processes necessary for these steps, so that post-project funding can be used primarily for formal certification and commercial scaling rather than late-stage R&D.

Planned fundraising activities
During and immediately after the project, DimensionLab plans to:

- Prepare for application to EIC Accelerator or similar EU instruments leveraging the Pre-accelerator results.
- Engage existing and new investors with updated traction metrics, regulatory roadmap and validated business model, targeting a follow-on seed/Series-A-equivalent round to finance certification and commercial rollout.
- Use project milestones (e.g. M6 mid-project review, pilot completion, regulatory-pack finalisation) as anchor points for investor updates, demo days and targeted roadshows.

By the end of the Pre-accelerator, the ambition is to have a technically robust, pilot-validated, regulatorily-mapped product, a clearly defined go-to-market strategy backed by real-world data, and an investor-ready company capable of absorbing and deploying significant follow-on capital to scale within Europe and globally.

DimensionLab expects to raise a **€300–500k bridge round** to support pilot activities and regulatory preparation. After this, and alongside the EIC Accelerator application, we plan a **€2M+ Seed/Series A round** to fund MDR certification and early commercial rollout.

The ideal investors for DimensionLab are specialised early-stage deep-tech and med-tech funds with experience in regulated software, digital surgery and AI-driven clinical workflows. We will prioritise investors who can provide not only capital but also strategic support in MDR/FDA pathways, hospital procurement, and commercial scaling in Europe and the US.

#§IMP-ACT-IA§#

3. Quality and efficiency of the implementation #@QUA-LIT-QL@# #@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.1 Quality and motivation of the team

DimensionLab is a deep-tech SME built around a compact, highly specialised team that combines **AI research, software engineering, biomedical applications and business/commercialisation expertise**. The founders have already demonstrated the ability to convince institutional investors and accelerators, having raised a pre-seed round of about **€1M from three European investors** and earlier support from a European accelerator, which is a strong external signal of quality and motivation to execute on a commercially ambitious roadmap. Investors own ~25% of the company through equivalent of SAFE convertible note.

CEO – Martin Muzelák (CEO, DimensionLab s.r.o.) - 12.5% stake in the company. Martin is a co-founder of DimensonLab and is responsible for overall strategy, execution and fundraising. He has **5+ years of experience in software engineering in multiple startup projects and also in enterprise environment**. He holds a **PhD in Cybernetics**, giving him a strong technical foundation to lead an AI-driven med-tech company and to communicate credibly with both engineers and clinicians. His track record includes being a **finalist at IBM Hack 2019** and achieving **2nd place at Falling Walls Lab 2022**, which illustrates his ability to turn complex technical ideas into compelling, innovation-oriented pitches. As CEO he drives partnerships with biomedical engineering companies, and ensures that the project remains aligned with a clear commercialisation and investment thesis.

CTO – Michal Takáč (CTO, DimensionLab s.r.o.) - 40% stake in the company. Michal is a co-founder of DimensionLab (formerly a CEO) responsible for technical strategy and fundraising. He brings **15+ years of software engineering and leadership experience** across SaaS and AI startups and has **co-founded four startups** to date. He holds a **PhD in Cybernetics** and was recognised as **Slovak Student Personality of the Year 2021** in the category of metallurgy, engineering and energy. As CTO he owns the technical roadmap and product architecture: translating state-of-the-art point-cloud diffusion research into a robust, maintainable codebase; enforcing quality and security practices; and making the technology deployable both on-premise and in the cloud. His combination of deep

technical skills and previous startup experience is central to turning the research into a scalable, commercially viable product.

CSO – Marek Ružička (CSO, DimensionLab s.r.o.). Marek is a **researcher in AI and robotics**, responsible for the scientific direction of the project. He leads the evaluation and adaptation of cutting-edge generative models, ensures that the implementation remains grounded in the latest literature, and designs experiments to rigorously validate geometry quality and robustness. As CSO he acts as the bridge between academic advances (including the point-cloud diffusion work on which we build) and industrial requirements such as manufacturability, uncertainty quantification and regulatory credibility.

CBO – Branislav Kršák (CBO, DimensionLab s.r.o.) - 12.5% stake in the company. Branislav is a co-founder of DimensionLab and contributes **extensive experience in complex decision-making, market planning and innovation management**. He has acted as an academic consultant in **more than 100 international projects** and as principal investigator in **27 projects**, many of them centred on the deployment of innovative ICT solutions in real-world sectors. Within DimensionLab he leads business strategy, market and competitor analysis, and the design of the commercial model. He also supports exploitation planning (including public-funding leverage) and facilitates the commercialisation of the research outputs towards hospitals, OEMs and investors.

Lead product & interaction design – Maroš Pekarík. Maroš has **8+ years of experience in software engineering and interaction design** and is a specialist in virtual spaces and interactive design. As a creative technologist active in robotics and immersive media art, he is uniquely positioned to design a 3D user interface that modelling teams actually want to use: intuitive, safe and efficient. He leads UX and visual design of the workstation and cloud UI, ensuring that the AI capabilities are presented in a way that fits existing clinical and manufacturing workflows.

CFO – to be appointed
A dedicated CFO is not yet in place, which is typical at this stage. Financial management is currently handled by the CEO with external accounting support. As the company moves into revenue and larger rounds (EIC Accelerator / private capital), DimensionLab plans to engage a **part-time/fractional CFO** with venture-backed med-tech experience, evolving this into a full-time role as the company scales.

Team motivation and incentives
To align the team with long-term value creation and attract senior talent, DimensionLab has an **Employee Stock Ownership Plan (ESOP) representing 10% of the company**, with standard vesting and cliff. This ensures key engineers, researchers and business staff share directly in the upside of successful commercialisation.

Overall, the team combines **deep AI/3D/cybernetics expertise, product-delivery experience, strong UX capabilities and proven business/commercialisation skills**. This is exactly what is needed to turn cutting-edge generative models into a clinically credible, user-friendly application with a compelling value proposition for customers and investors.

3.2 KPIs, milestones and risks

We define clear performance indicators for the AI model (accuracy, robustness, QA pass rates), for the integrated prototype (usability, reliability, time-to-result), and for business-model maturation (number of customer-discovery meetings, pricing tests, LOIs secured, investor interactions, and partnership engagements). These KPIs allow us to track progress toward both technical validation (TRL4→TRL5) and early commercial validation.

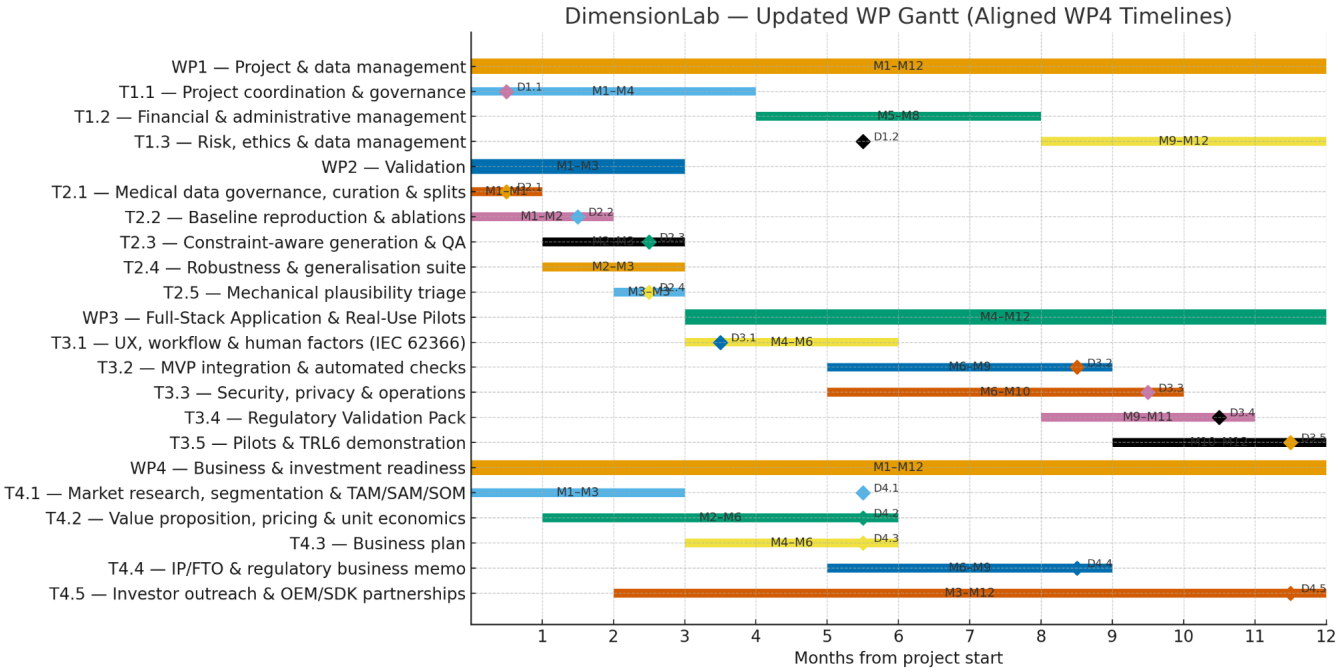
We also identify the critical technical and market risks that may affect progress—such as data variability, model generalisation limits, regulatory changes, procurement barriers, or delays in pilot onboarding—and describe targeted mitigation actions for each. All KPIs, milestones, and risks are summarised in tables 3.3d and 3.3e, ensuring transparent monitoring throughout the project.

3.3 Work plan and resources #@WRK-PLA-WP@#

The following work plan outlines the structure, timing and interdependencies of all activities across the 12-month Pre-Accelerator project. Each work package is sequenced to ensure a smooth progression from technical

validation (WP2) to prototype integration and pilots (WP3), supported throughout by project management (WP1) and business/investment readiness activities (WP4). The GANTT chart below provides a clear overview of the timeline, milestones and resource allocation across the project.

GANTT diagram



#§CON-SOR-CS§# #§PRJ-MGT-PM§

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Table 3.3a: List of work packages

Work package No	Work Package Title	Lead Participant No	Lead Participant Short Name	First name and family name of the Work Package Leader	Gender of Work Package Leader	Person-Months	Start Month	End month
WP1	Project and data management	1	DimensionLab	Branislav Kršák	Male	3	1	12
WP2	Validation	1	DimensionLab	Marek Ružička	Male	18	1	3
WP3	Full-Stack Application & Real-Use Pilots	1	DimensionLab	Michal Takáč	Male	46.50	4	12
WP4	Business and investment readiness	1	DimensionLab	Martin Muzelák	Male	22	1	12
						89.50		

Table 3.3b: Work package description

Work package number	WP1
Work package title	Project and data management
Objectives This WP ensures effective coordination, governance, and technical oversight of all project activities and outputs. It provides robust financial, administrative, and risk management to support smooth implementation. It also guarantees full compliance with the Grant Agreement, including EU requirements on ethics, data protection, and reporting.	

Description of work

Task 1.1 Project coordination & governance -> D1.1

This task ensures strategic and day-to-day coordination of project activities, including scheduling and monitoring of technical progress. The Project Coordinator will oversee timely delivery of milestones and outputs, lead internal reviews, and maintain communication with the European Commission. Collaboration with external stakeholders will be fostered to align the project with market needs and exploitation goals. A *Project Management Handbook* will guide internal workflows and decision-making. This ensures coherence, transparency, and efficiency across the consortium.

Task 1.2 Financial and administrative management -> D1.1

This task ensures robust financial and administrative implementation in full alignment with the Grant Agreement and EU funding rules (lump sums). It covers budget monitoring, co-funding tracking, and accurate allocation of expenditures within the limits of the lump sums. The task includes the preparation of reports and documentation for reviews. It also manages contracts, invoices, and internal policy documents essential for project delivery. The goal is to maintain transparency, compliance, and efficient resource use.

Task 1.3 Risk, ethics and data management -> D1.2

This task focuses on proactive risk management, covering technical, operational, and strategic dimensions. A living Risk Register will be maintained and updated with mitigation actions as the project evolves. The Project Coordinator will oversee ethical compliance in line with EU standards. A *Data Management Plan* will be developed and regularly reviewed to ensure secure, FAIR, and GDPR-compliant handling of all project data. These measures aim to safeguard project integrity and compliance.

Work package number	WP2
Work package title	Validation

Objectives

Establish a robust, reproducible diffusion pipeline for automatic cranial implant generation, with quantified performance and robustness suitable for relevant-environment use (≥TRL5). We will reproduce the core point-cloud diffusion approach, verify it on representative EU datasets, and harden it with constraint-aware post-processing suitable for clinical modeling workflows. Emphasis is on measurable geometry fidelity, watertightness, and time-to-result that meet pre-defined acceptance thresholds. Outputs will be documented in SOPs, dataset cards, and a validation report that can be reviewed by clinical partners and investors.

Description of work

Task 2.1:Medical data governance, curation & splits -> D2.1

We will de-identify CT DICOM per HIPAA 45 CFR §164.514 and execute BAAs with pilot partners before any data transfer. DICOM will be converted to meshes with consistent windowing and segmentation protocols, and we will curate defect typologies (trauma, decompressive craniectomy, post-tumor) that reflect real clinical variability. Stratified train/validation/test splits will be designed to avoid patient leakage and to enable cross-site generalization analysis. A formal **Dataset Card** and **Data Management Plan** will capture provenance, inclusion/exclusion criteria, and access controls. Traceability from case → masks → meshes will be maintained to support reproducibility and later regulatory evidence.

Task 2.2 — Baseline reproduction & ablations -> D2.2

We will reproduce the point-cloud diffusion baseline, then systematically ablate point sampling strategies, denoising schedules, conditioning inputs (defect mask, symmetry planes), and guidance strengths. A comparative evaluation of mesh reconstruction (Poisson, ball-pivoting, implicit SDF) will focus on rim fidelity, watertightness, and surface smoothness required for milling/3D-printing. Each ablation will be logged with seeds and configs to ensure determinism and enable later audits. Results will inform the default training/inference recipe used downstream in WP3.

Task 2.3 — Constraint-aware generation & implant Quality Assurance (QA) -> D2.3

We will add constraints for **minimum/nominal thickness**, **edge blending** at the skull rim, and **templated screw-hole patterns** that match common surgical fixation techniques. Automated checks will verify watertightness, manifoldness, and collision/overlap with surrounding bone to prevent downstream corrections. Geometry-aware losses (e.g., Chamfer, normal consistency) plus boundary penalties will bias the generator

toward rim-conforming, fabrication-ready shapes. A “red-flag” system will block export when violations exceed thresholds and route cases to human review.

Task 2.4 — Robustness & generalisation suite -> D2.4

We will stress-test the pipeline across scanners, protocols, and defect morphologies, including partial scans, simulated noise, and HU-windowing perturbations. Unseen defect shapes and adversarial missing-data tests will quantify generalization and failure surfaces. An outlier detector will trigger human-in-the-loop checks when uncertainty or data shift is detected, reducing the likelihood of silent failures. These tests will be repeatable and bundled into the CI pipeline.

Task 2.5 — Mechanical plausibility triage -> D2.5

Lightweight, surrogate FEA checks will estimate local thickness, curvature, and stress “hot-spots” as a screening tool before expert sign-off. The triage will not claim clinical accuracy but will reliably flag suspicious geometries for closer inspection. This supports our **design-assist** intended use and reduces rework in later pilot phases. The triage thresholds will be calibrated against retrospective cases and expert feedback.

Work package number	WP3
Work package title	Full-Stack Application & Real-Use Pilots

Objectives

Create a **local & cloud** application with interactive 3D UI, integrated diffusion pipeline, automated QA, and secure operations, culminating in pilots that demonstrate **TRL5 minimum** and **TRL6 as a stretch**. The product will be designed as a **design-assist** tool, keeping intended-use claims strictly within planning/design support to limit regulatory burden at this stage. The mid-project review in **M6** will assess investor readiness and Fast-Track eligibility. By M12, we aim for operational evidence that the tool fits daily modeling workflows.

Description of work

Task 3.1 — UX, workflow & human factors -> D3.1

We will perform contextual inquiries and task analyses with modeling teams to define critical tasks and error-prone steps. **IEC 62366** will guide the usability plan, including formative studies and risk-controlled UI patterns. Early prototypes will be evaluated using SUS and task-completion metrics to lock navigation and tool ergonomics. Findings will directly inform guardrails, prompts, and defaults that prevent unsafe edits.

Task 3.2 — MVP integration & automated checks -> D3.2

We will embed WP2’s model-of-record and constraints into an end-to-end pipeline: **DICOM** → **mask/mesh** → **diffusion** → **QA** → **edit** → **export**. Automated checks (rim fit, thickness, curvature, collision) will run in the background and surface actionable messages with one-click fixes where possible. A baseline-comparison view will let users compare AI implants with historical manual designs to build trust. The MVP will target predictable performance on commodity GPUs and document hardware profiles.

Task 3.3 — Security, privacy & operations -> D3.3

HIPAA controls—de-identification, access logging, and least-privilege roles—will be implemented in both local and cloud deployments. **MFA** and encryption **in transit/at rest** will be enforced, with a key-management and incident-response playbook appropriate for SMEs. We will monitor **2025 HIPAA security posture** updates and apply relevant changes (e.g., MFA hardening) before pilots. BAAs will be executed with pilot sites, installers hardened, and a GDPR DPIA completed to address EU data protection.

Task 3.4 — Regulatory Validation Pack -> D3.4

We will map the solution to **FDA 2023 Modeling & Simulation (M&S) credibility** guidance, aligned with **ASME V&V 40** (context of use, credibility goals, verification/validation/uncertainty). **IEC 62304** (Class **B**) processes will cover requirements, architecture, unit/integration tests, anomaly tracking, and release notes; usability under **IEC 62366** and QMS-lite aligned to **ISO 13485** will ensure traceability. A **risk traceability matrix** (ISO 14971 principles) will link hazards → mitigations → tests → evidence. The pack will define intended-use/claims that remain design-assist, deferring certification to the Accelerator phase.

Task 3.5 — Pilots & TRL6 demonstration -> D3.5

We will run 2–3 EU pilots with hospital 3D labs or modeling service providers, capturing end-to-end case time, edit minutes, acceptance rate, and failure modes. Human-factors observations will test whether guardrails

reduce errors and rework. Operational KPIs (latency, crash-free sessions, security posture) will be monitored to prove real-world reliability. We will prepare an investor pack and **EIC Accelerator** short-form, and the **M6** mid-project review will assess Fast-Track readiness.

Work package number	WP4
Work package title	Business and investment readiness

Objectives

Build and validate a credible business model and go-to-market pathway for AI-assisted cranial implant design, anchored in real feedback from hospitals, modelling labs and implant OEMs. WP4 will translate technical milestones into a clear value proposition, pricing and unit economics, and a concrete commercial roadmap, resulting in LOIs, pilot contracts and an investor-ready story. By the end of the project, the company should be demonstrably investment-ready (for EIC Accelerator and private capital), with a tested business model, initial commercial traction and a prepared data room.

Description of work

Task 4.1 — Market research, segmentation & TAM/SAM/SOM analysis validation→ D4.1
We have already prepared a first version of our TAM/SAM/SOM analysis for cranial implant markets. During the project, we will validate and refine this analysis with real pilot clients (hospital 3D labs, modelling services, implant OEMs), using their volumes, pricing and workflow data to adjust our assumptions. This iterative process will sharpen our segmentation and confirm which niches are most attractive for early commercial focus. The validated TAM/SAM/SOM figures will feed directly into the business plan and financial model, underpinning our go-to-market strategy and investor narrative.

Task 4.2 — Value proposition, pricing & unit economics validation → D4.2
We have already created an initial Value Proposition Canvas and Business Model Canvas for our implant-design software, based on early interviews and desk research. Within the project, we will iterate these canvases together with pilot clients, testing and refining the value hypotheses. In parallel, we will validate unit economics beyond software licensing fees, including per-implant internal costs (labour, infrastructure, support) and potential revenue-sharing models, to ensure that our pricing is sustainable for both us and our customers.

Task 4.3 — Business plan → D4.3, D4.7
This task synthesises all commercial, financial, regulatory and market insights developed in Tasks 4.1–4.2 and integrates them into a coherent, investor-grade business plan. It will include a refined Business Model Canvas, a validated go-to-market strategy, pricing model, customer segmentation, competitive positioning, regulatory implications for commercialisation, and a detailed financial model with 3–5-year projections. The business plan will also formalise the commercial rollout sequence informed by WP3 pilot activities—covering early-adopter profiles, sales funnel design, support requirements, and scale-up assumptions. It will define strategic partner pathways (OEMs, modelling labs, implant 3D-printing ecosystem vendors) and outline priority geographies based on reimbursement, regulatory timing, and market readiness.

Task 4.4 — IP/FTO, regulatory business memo & commercial risk analysis → D4.4
In parallel with technical and regulatory work in WP2–WP3, this task will consolidate the business-side implications of IP and regulation. We will work with legal and regulatory advisors to perform a focused FTO scan around PSCI design software and related CAD/AI patents, and define where patents vs. trade secrets are most appropriate. We will prepare a concise business-oriented memo summarising: intended claims and design-assist positioning, regulatory pathway options (EU MDR, FDA), implications for sales and deployment (SaaS vs on-prem), and key commercial risks (e.g. classification changes, reimbursement issues). This memo will be used both internally (roadmapping) and externally (investor discussions, strategic partner talks).

Task 4.5 — Investor outreach, LOIs & OEM/SDK partnerships → D4.5, D4.8
This task focuses on converting technical and pilot progress into concrete commercial and investment signals. We will work with pilot sites from WP3 to secure **Letters of Intent (LOIs)** and, where feasible, move towards **first paid PoCs/pilots** within or shortly after the project. In parallel, we will reach out to implant OEMs, hospital-IT vendors and 3D-printing ecosystem players (hardware and software) to explore OEM/SDK partnerships and integration paths. For investors, we will prepare and refine a data room (deck, financial model, traction KPIs, regulatory roadmap, IP status) and run targeted outreach to EIC Accelerator evaluators and specialised deep-tech/med-tech funds.

Table 3.3c: List of Deliverables

Number	Deliverable name	Short description	WP No	Short name of lead participant	Type	Dissemination level	Delivery date (in months)
D1.1	Project management handbook	Governance, Roles, Procedures, Risk management & quality assurance	WP1	DimensionLab	R	SEN	1
D1.2	Data management plan	First version of the DMP	WP1	DimensionLab	DMP	SEN	6
D2.1	Dataset Card & SOPs	Dataset governance, SOPs, and site/data onboarding criteria	WP2	DimensionLab	R	SEN	1
D2.2	Reproducible Baselines & Ablations	Versioned code/assets enabling safe, reproducible benchmarking	WP2	DimensionLab	R	SEN	2
D2.3	Constraint-aware Generator & QA Toolkit	Modular generator with automatic QA and WP3 integration tools	WP2	DimensionLab	R	SEN	3
D2.4	Bench Validation Report	TRL5 validation report with performance metrics and integration readiness	WP2	DimensionLab	R	SEN	3
D3.1	High-fidelity UI Prototypes & Usability Report	Interactive prototypes and UX study results with mitigation roadmap	WP3	DimensionLab	R	SEN	4
D3.2	Local & Cloud MVP (v1) with QA/Exports	MVP release with QA, STL/STEP export, and deployment profiles	WP3	DimensionLab	R	SEN	9
D3.3	Security & Privacy Controls Pack	MFA, encryption, access logs, incident drill, DPIA summary	WP3	DimensionLab	R	SEN	10
D3.4	Regulatory Validation Pack	Regulatory traceability, intended use, claims, and evidence gaps	WP3	DimensionLab	R	SEN	11
D3.5	Pilot Report & TRL5/6 + Readiness Pack	Pilot results, TRL evidence, investor materials, go/no-go criteria.	WP3	DimensionLab	R	SEN	12
D4.1	Market research, segmentation & TAM/SAM/SOM analysis validation	Validated market segmentation and TAM/SAM/SOM model based on real data from pilot hospitals and implant OEMs; includes updated assumptions for volumes, pricing and workflows.	WP4	DimensionLab	R	SEN	6
D4.2	Value proposition, pricing & unit economics validation	Evidence-based validation of value proposition, pricing strategy and unit economics, including refined Value Proposition Canvas, Business Model Canvas inputs and per-implant cost structures.	WP4	DimensionLab	R	SEN	6
D4.3	Draft Business plan	Preliminary, investor-ready business plan integrating validated market insights,	WP4	DimensionLab	R	SEN	6

		competitive analysis, go-to-market strategy, pricing model and financial projections					
D4.4	IP/FTO, regulatory business memo & commercial risk analysis	Consolidated business-focused IP/FTO assessment, regulatory pathway options (EU MDR & FDA), deployment implications (SaaS/on-prem) and commercial risk analysis.	WP4	DimensionLab	R	SEN	6
D4.5	Investor outreach, LOIs & OEM/SDK partnership pack	Investor & EIC Accelerator readiness pack including pitch deck, financial model, LOIs from pilot sites, OEM/SDK partnership roadmap and supporting commercial evidence.	WP4	DimensionLab	R	SEN	12
D4.7	Final Business plan	Final, investor-ready business plan integrating validated market insights, competitive analysis, go-to-market strategy, pricing model and financial projections	WP4	DimensionLab	R	SEN	12
D4.8	Plan for dissemination and exploitation	Plan for dissemination and exploitation including communication activities	WP4	DimensionLab	R	SEN	3

Table 3.3d: List of milestones

Milestone No.	Milestone name	Related WP(s)	Due date (in month)	Means of verification (KPI)
M1	TRL4 concluded	WP2	3	Performance on held-out, relevant-environment data meets thresholds, with SOPs and acceptance criteria approved by advisors. The model-of-record, constraints, and QA toolkit are packaged for WP3 integration. Known limitations and safe fallbacks are documented to support risk-managed pilots.
M2	TRL5 concluded	WP3	12	Pilots show users complete cases under SOP with acceptance thresholds met, demonstrating TRL5 in relevant environments. Where sites operate under near-clinical conditions with documented reliability and human-factors success, we will enter TRL6. The Regulatory Pack and investor materials will be complete to support an EIC Accelerator submission.
M3	Business Readiness	WP4	12	Investment and exploitation readiness achieved. Verification: At least one LOI or pilot contract signed (showing commercial traction), investor data room populated and shared with prospects, and a follow-on funding application (EIC Accelerator or equivalent) submitted.

Table 3.3e: Critical risks for implementation #@RSK-MGT-RM@#

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	WPs	Proposed risk-mitigation measures
Insufficient data diversity	WP2	Expand dataset via synthetic defects and domain

(Likelihood: Medium; Severity: High) Homogeneous patient data or scanner setups may reduce generalizability and limit robustness of model validation across sites.		augmentation. Onboard diverse sites early. Use cross-site generalization tests as integration gates.
Site/scanner overfitting and drift *(Likelihood: Medium; Severity: Medium)* Model may overfit to site-specific patterns, causing performance drops in new environments.	WP2	Apply leave-one-site-out validation. Normalize inputs across scanners. Track site-wise metrics and adapt training if drift is observed.
Mesh defects in outputs *(Likelihood: Medium; Severity: High)* Non-watertight or invalid geometry may block downstream printing and delay pilot integration.	WP2	Use mesh repair and automatic topology checks. Block faulty exports. Include QA review and feedback into model tuning.
Cloud inference latency or cost bottlenecks *(Likelihood: Medium; Severity: High)* Delays or high costs in cloud-based inference may impair pilot usability or scalability.	WP3	Enable mixed precision, on-demand scaling, and smart caching. Provide on-prem fallback for sensitive sites. Use telemetry to detect regressions early.
Usability gaps in edge cases *(Likelihood: Medium; Severity: Medium)* Edge-case interactions may confuse users or reduce task efficiency.	WP3	Run synthetic HFE tests and log task-level UX data. Implement guided correction flows. Prioritize fixes based on impact and frequency.
Regulatory scope creep *(Likelihood: Low; Severity: High)* New features may unintentionally expand regulatory classification beyond current claims.	WP3	Limit claims to design-assist. Use regulatory checklist to screen features. Flag high-risk items for post-project roadmap only.
Dependency on technical milestones for commercial traction *(Likelihood: Medium; Severity: Medium)* If core technical implementation slips, WP4 may be unable to secure timely Letters of Intent (LOIs) or demonstrate sufficient business value to early adopters.	WP4	Closely couple WP4 roadmap to WP2–WP3 milestones; use a shared milestone tracker and regular cross-WP reviews. Prioritise a narrow, pilot-ready feature set ("minimum viable product") rather than broad functionality, so that value demos and LOIs can start as soon as TRL5 is reached, even if later features slip.
Delay of first paid PoC/pilots due to under-resourcing *(Likelihood: Medium; Severity: Medium)* A 2-month delay in development, caused by insufficient technical headcount or funding, could prevent execution of the first paid PoCs/pilots within the project period and weaken the commercial case.	WP4	Secure adequate runway early (bridge funding, reallocation of internal budget) to hire or contract the required engineers; define "pilot-critical" features and protect them from scope creep. Use phased pilot agreements that allow initial unpaid evaluation followed by paid extensions, so that commercial signals (and revenue) can be captured even if full-feature delivery shifts slightly.

#§RSK-MGT-RM§#

Table 3.3f: Summary of staff effort

	WP1	WP2	WP3	WP4	Total Person-Months
1/DimensionLab	3.0	18.0	46.0	22.0	**89.5**

Table 3.3g: 'Subcontracting costs' items: NOT RELEVANT

Table 3.3h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

Please complete the table below if the purchase costs (i.e. the sum of the costs for 'travel and subsistence', 'equipment', and 'other goods, works and services') exceeds 15% of the personnel costs for that participant (according to the budget table in proposal part A). The record must list cost items in order of costs and starting with the largest cost item, up to the level that the remaining costs are below 15% of personnel costs. Please be aware that CFS and audit costs are not needed and therefore not eligible in lump sum.

1/DimensionLab		
	Cost (€)	**Justification**
Travel and subsistence	18 000	Travel is required for coordination meetings with pilot sites, usability evaluations, workflow validation, investor outreach, SARIO Missions like Indianapolis, EU Business Hub Missions like KIMES 2026 and OEM/SDK partnership discussions.
Equipment	0	
Other goods, works and services	118 000	Covers EU-based cloud GPU compute for training, robustness testing and pilot inference (H100 instances); specialised external services for IP/FTO analysis, regulatory pathway (MDR/FDA) support, market analysis, and business planning essential for WP4 investment readiness.
Remaining purchase costs (<15% of pers. Costs)		
Total	118 000	

Table 3.3i: 'Other costs categories' items (e.g. internally invoiced goods and services): NOT RELEVANT
Table 3.3j: 'In-kind contributions' provided by third parties: NOT RELEVANT

#§QUA-LIT-QL§# #§WRK-PLA-WP§#

Seal of Excellence

If my application is evaluated as meeting all the criteria and thresholds for funding but remains unfunded due to lack of budget:

I, Branislav Kršák, the coordinator of this proposal, consent to share with our relevant National Contact Point, EEN Member and other funding public organisations from the European Union, EU member states and Horizon Europe associate countries (including regional innovation organisations, European Structural and Investment Funds (ESIF) Managing authorities etc.), the following data:

basic information about my proposal (proposal acronym, title, abstract, amount requests and the evaluation result);
personal data (applicant's contact details, e.g. email, name);

Data will be made available subject to confidentiality obligations agreed with the NCPs, EEN members and relevant funding bodies. Please note that without this confirmation, no Seal of Excellence can be awarded.